

Functions I

1. Find the domain and range of the following:

$$y = 5x^2 + 2, \quad y = \sqrt{x^2 - 8x + 15}, \quad y = \tan x, \quad y = \frac{1}{\cos x}, \quad y = \frac{2x + 7}{4x - 5}.$$

2. Let $f(x) = 2x^2 - 5x + 9$. Compute

$$f(2x), \quad 3f(x), \quad f(ix), \quad 2f\left(\frac{1}{3}x - 5\right).$$

3. Let $f(x) = 2x^2 + 7x - 15$ and $g(x) = x^2 - 1$. Compute

$$f + g, \quad f - g, \quad f \circ g, \quad f(g(1)), \quad (g/f)(1).$$

4. Let $F(x) = 3x$. **(a)** Show that $F(x_1 + x_2) = F(x_1) + F(x_2)$. **(b)** Show that $F(cx_1) = cF(x_1)$. **(c)** Show that in general, if $L(x) = mx$, then $L(x_1 + x_2) = L(x_1) + L(x_2)$ and $L(cx_1) = cL(x_1)$.

5. Let $f(x) = x^2$. Graph

$$f(x), \quad f(x) - 2, \quad f(x + 2), \quad -2f(x + 2), \quad \frac{1}{2}f(x - 2) + 2$$

6. Suppose $x = 2$ is the only root of a function f . Determine the roots of the following functions:

$$-f(x), \quad f(f(x)), \quad f(x^2), \quad 3f(x), \quad f(x^3 - 2), \quad f\left(\frac{x}{2}\right).$$

If the roots cannot be determined from the given information, write “UND”.

7. Find the average rates of change on the interval $[3, 12]$ for

$$f(x) = -3x + 2, \quad f(x) = x^2 - 11x + 24, \quad f(x) = \sqrt{x^3 - x^2 + 12}.$$

8. **(a)** A function f is said to be *even* if $f(-x) = f(x)$ for all x in its domain. Show that if (x, y) is a point on the graph of an even function f , then so is $(-x, y)$. **(b)** A function f is said to be *odd* if $f(-x) = -f(x)$ for all x in its domain. Show that if (x, y) is a point on the graph of an odd function f , then so is $(-x, -y)$. **(c)** Is each function (with domain $\mathbb{R}$) even, odd, or neither?

$$a(x) = x^3, \quad b(x) = x^2 - 2, \quad c(x) = x^3 + x^2, \quad d(x) = \sin(x), \quad e(x) = |x|$$

9. Suppose the position (height) of a falling object measured in feet follows the equation

$$h(t) = -16t^2 + 144$$

where t is measured in seconds. **(a)** When does the object hit the ground? **(b)** Find a general expression for the average rate of change $\Delta h / \Delta t$ over the interval $[1/2, 1/2 + x]$. **(c)** What is a physical interpretation of the expression you found in part (b)? **(d)** Evaluate your expression as x tends to 0. What is a physical interpretation of the resulting expression?

10. Definition. For a function F and a fixed input x , define the *difference quotient*

$$A(F, h, x) = \frac{F(x+h) - F(x)}{h}, \quad h > 0.$$

(a) Let f and g be functions and c a real constant. Recall that $(f+g)(x) = f(x) + g(x)$ and $(cf)(x) = c \cdot f(x)$. Show that for any x and any $h > 0$:

$$\begin{aligned} A(f+g, h, x) &= A(f, h, x) + A(g, h, x), \\ A(cf, h, x) &= c \cdot A(f, h, x). \end{aligned}$$

(b) Let $f(x) = x^2$. Compute $A(f, h, x)$ and simplify your answer as much as possible. What value does $A(f, h, x)$ approach as h gets closer and closer to 0?